Facts Table Report

Setup

```r
library(DBI)
library(RMySQL)


DB_CERT <- "-----BEGIN CERTIFICATE-----
MIIERDCCAqygAwIBAgIUJVlW/RqVEGZ6kc/3YW/FWoZc9AwwDQYJKoZIhvcNAQEM
BQAwOjE4MDYGA1UEAwwvYjFmNTYxNTMtYzkzYiOOOGQzLTk2M2QtM2ZjN2Q2NDg4
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MTgwNgYDVQQDDC9iMWY1NjE1My1jOTNiLTQ4ZDMtOTYzZCOzZmM3ZDYOODg1OTMg
UHJvamVjdCBDQTCCAaIwDQYJKoZIhvcNAQEBBQADggGPADCCAYoCggGBAKGs4Eym
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mO2lzvQ1ayr2u3Jpca1JgFkf5OWUZe17XsgB/S/Uid/AVX+DuDnPn6User3MaRdP
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59u7us81efUbwG1E572BiKJ/SS2lGWz8Y5B1xNq6EVdN3IrCU+YAVDqWqtvRgSMz
4aTVExrVhqOBJ8uVqpXeEbwt4MmBN33FAnhztLX+nMFin3UpcqaWJ+Bf/khZEZsI
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7JrER9UDhHUhXWsYlgPEck+c4SE+boIOdLRPmCbIHVPGupHwjixxYmKAnp2uTtC4
KtE7cMvj3dGttYOVC74SW6tOLdyzBHiHqC5cXeJ/Hpdjkpz/nsTu/CNSbwIDAQAB
oOIwQDAdBgNVHQ4EFgQUGTJ5NadKR2JP86q2RxBSuaAXxcswEgYDVROTAQH/BAgw
BgEB/wIBADALBgNVHQ8EBAMCAQYwDQYJKoZIhvcNAQEMBQADggGBADDCcIIQSqmf
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Hm8TN6vcq+bhwO7/SPjU2/27A4zpFiE8oNpLF8e1R6geth4ChO3pdxLttMe9Ktjq
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T6l6af0c72tSauqnwM79SEsmg8VNScwtKcmv/06VaTTGLUhpdSaoNOsce2D/AiND
+xG2mvnX+pUpEGRTxChhgX23OSjOdm/yEI6XCPGNT4Fi4+N3VLxFKBINOTWa8WrL
nor6EjVCHn+zJJeaE++cK+fLoeMnIehpsKozJotfFm55G6tyhtRzYA==
-----END CERTIFICATE-----
"

mart <- dbConnect(MySQL(),
                  user = "avnadmin",
                  password = "AVNS_egLExJvgSpso0YfSzez",
                  dbname = "defaultdb",
                  host = "practicum-i-cs5200-harrisj-jazmynrh12-9958.b.aivencloud.com",
                  port = 19929,
                  sslmode = "require",
                  sslcert = DB_CERT)
```

Expense Analysis Report

Intro

Heracletes Advisors PC is interested in keeping track of the business expenses that they accrued in order to bill clients. Expenses are accumulated from employee transactions that take place during travel for a given project. This expense reporting system allows employees to easily upload the transactions that they have garnered for a given credit card merchant, month, and year. The report system will extract information to produce different expense facts. This report will demonstrate an expense analysis from this reporting system.

Expense Facts Report

GENERAL TABLE

```
sql <- "
SELECT *
FROM ExpenseFacts f
LIMIT 5"

dbGetQuery(mart, sql)
```

```
##   ExpenseFactsID DateDimID ClientProjectDimID CategoryDimID CurrencyDimID Year
## 1              6      3001                  6             1             1 2025
## 2              7      4001                  7             9             1 2025
## 3              8      3002                  9             4             1 2025
## 4             10      3001                  1             1             1 2025
## 5             13      2005                  2             3             1 2025
##   Quarter TotalAmount AverageAmount
## 1       4      565.65        282.83
## 2       4      660.14        132.03
## 3       4      233.76        233.76
## 4       4      793.23        198.31
## 5       3      144.58        144.58
```

Question 1: What are the total amount and average of each transaction spent (in USD) for each client in all time? (sorted by ClientName)

```
sql <- "
    SELECT
    cp.ClientName,
    SUM(f.TotalAmount * cu.USExchangeRate) AS TotalSpentUSD,
    AVG(f.AverageAmount * cu.USExchangeRate) AS AvgTransactionUSD
    FROM ExpenseFacts f
    JOIN ClientProjectDim cp ON f.ClientProjectDimID = cp.ClientProjectDimID
    JOIN CurrencyDim cu ON f.CurrencyDimID = cu.CurrencyDimID
    GROUP BY cp.ClientName
    ORDER BY TotalSpentUSD DESC
    LIMIT 5"

dbGetQuery(mart, sql)
```

```
##                     ClientName TotalSpentUSD AvgTransactionUSD
## 1                    Rio Tinto      306568.7          136.1857
## 2                     Glencore      303183.3          136.9982
## 3 Southern Copper Corporation      299674.1          138.5796
## 4                    BHP Group      299476.8          136.8582
## 5                      Vale SA      297855.1          137.2442
```

Time-Based

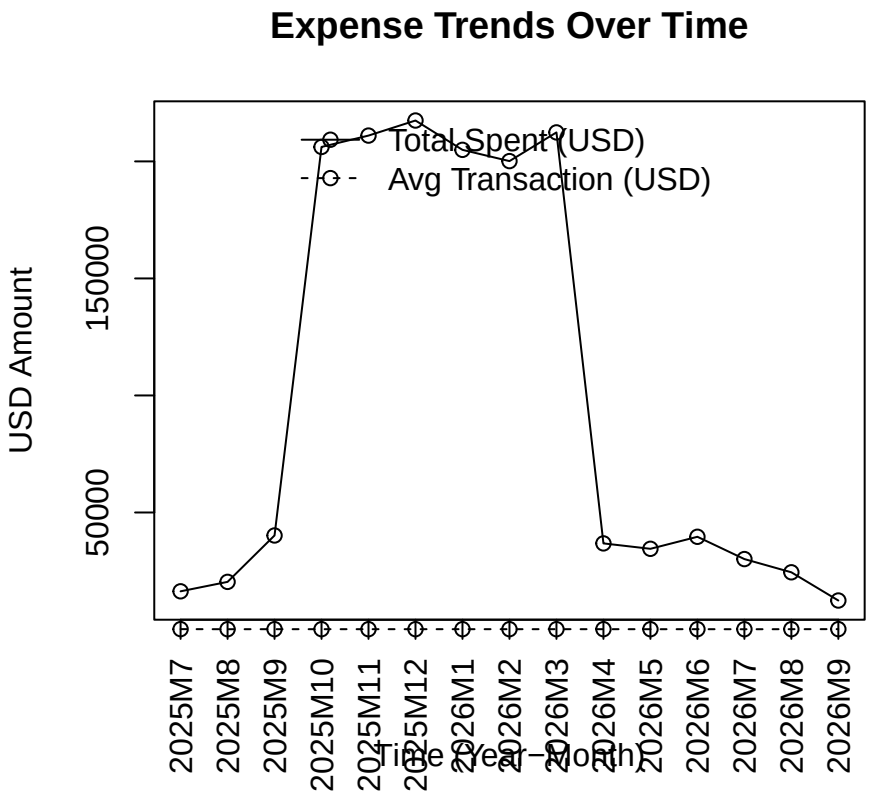
Question 2: What is the trend of the total and average expense over time (by year and Month?

```
sql <- "
SELECT
    f.Year,
    d.Month,
    SUM(f.TotalAmount * cu.USExchangeRate) AS TotalSpentUSD,
    AVG(f.AverageAmount * cu.USExchangeRate) AS AvgTransactionUSD
FROM ExpenseFacts f
JOIN DateDim d ON f.DateDimID = d.DateDimID
JOIN CurrencyDim cu ON f.CurrencyDimID = cu.CurrencyDimID
GROUP BY f.Year, d.Month
ORDER BY f.Year, d.Month
"

df <- dbGetQuery(mart, sql)
```

GRAPH

```
# Create a time label (Year-M)
df$Time <- paste(df$Year, "M", df$Month, sep="")
df$Time <- factor(df$Time, levels = df$Time)

# Expand right margin
par(mar = c(5, 4, 4, 10), xpd = TRUE)

# Plot Total and Average on same graph
plot(
  x = seq_along(df$Time),
  y = df$TotalSpentUSD,
  type = "o",
  xaxt = "n",
  xlab = "Time (Year-Month)",
  ylab = "USD Amount",
  main = "Expense Trends Over Time"
)

# Add x-axis labels
axis(1, at = seq_along(df$Time), labels = df$Time, las = 2)

# Add Average line
lines(
  x = seq_along(df$Time),
  y = df$AvgTransactionUSD,
  type = "o",
  lty = 2
)

# Add legend
legend(
  "top",
  inset = c(-0.25, 0),
  legend = c("Total Spent (USD)", "Avg Transaction (USD)"),
  lty = c(1, 2),
```

```
  pch = 1,
  bty = "n"
)
```


Expense Trends Over Time
Total Spent (USD)
Avg Transaction (USD)
USD Amount
50000
150000
2025M7
2025M8
2025M9
2025M10
2025M11
2025M12
2026M1
2026M2
2026M3
2026M4
2026M5
2026M6
2026M7
2026M8
2026M9
Time (Year−Month)

Question 3: Which quarter has the highest spending overall across all clients?

```r
sql <- "
SELECT
    f.Year,
    f.Quarter,
    SUM(f.TotalAmount * cu.USExchangeRate) AS TotalSpentUSD
FROM ExpenseFacts f
JOIN CurrencyDim cu ON f.CurrencyDimID = cu.CurrencyDimID
GROUP BY f.Year, f.Quarter
ORDER BY TotalSpentUSD DESC
LIMIT 1"

dbGetQuery(mart, sql)
```

```
##   Year Quarter TotalSpentUSD
## 1 2025       4      634672.5
```

Client-Based

Question 4: Which clients/projects have the highest total spending across all time?

```r
sql <- "
SELECT
    cp.ClientName,
    cp.ProjectName,
    SUM(f.TotalAmount * cu.USExchangeRate) AS TotalSpentUSD
FROM ExpenseFacts f
JOIN ClientProjectDim cp ON f.ClientProjectDimID = cp.ClientProjectDimID
JOIN CurrencyDim cu ON f.CurrencyDimID = cu.CurrencyDimID
GROUP BY cp.ClientName, cp.ProjectName
ORDER BY TotalSpentUSD DESC
LIMIT 5"

dbGetQuery(mart, sql)
```

```
##                    ClientName                     ProjectName TotalSpentUSD
## 1                   Rio Tinto     Rio Tinto Compliance Review      153551.9
## 2                   Rio Tinto       Rio Tinto Site Assessment      153016.8
## 3                    Glencore        Glencore Site Assessment      151779.3
## 4                    Glencore      Glencore Compliance Review      151404.1
## 5 Southern Copper Corporation Southern Copper Site Assessment      150785.2
```

Category-Based

Question 6: Which expense categories have the highest total spent?

```
sql <- "
SELECT
    cp.ClientName,
    cp.ProjectName,
    SUM(f.TotalAmount * cu.USExchangeRate) AS TotalSpentUSD
FROM ExpenseFacts f
JOIN ClientProjectDim cp ON f.ClientProjectDimID = cp.ClientProjectDimID
JOIN CurrencyDim cu ON f.CurrencyDimID = cu.CurrencyDimID
GROUP BY cp.ClientName, cp.ProjectName
ORDER BY TotalSpentUSD DESC
LIMIT 5"

dbGetQuery(mart, sql)
```

```
##                     ClientName                     ProjectName TotalSpentUSD
## 1                   Rio Tinto     Rio Tinto Compliance Review      153551.9
## 2                   Rio Tinto       Rio Tinto Site Assessment      153016.8
## 3                    Glencore        Glencore Site Assessment      151779.3
## 4                    Glencore      Glencore Compliance Review      151404.1
## 5 Southern Copper Corporation Southern Copper Site Assessment      150785.2
```

Question 7: What categories dominate spending for each client?

```
sql <- "
SELECT *
FROM (
    SELECT
        cp.ClientName,
        c.CategoryName,
        SUM(f.TotalAmount * cu.USExchangeRate) AS TotalSpentUSD,
        RANK() OVER (
            PARTITION BY cp.ClientName
            ORDER BY SUM(f.TotalAmount * cu.USExchangeRate) DESC
        ) AS rnk
    FROM ExpenseFacts f
    JOIN ClientProjectDim cp ON f.ClientProjectDimID = cp.ClientProjectDimID
    JOIN CategoryDim c ON f.CategoryDimID = c.CategoryDimID
    JOIN CurrencyDim cu ON f.CurrencyDimID = cu.CurrencyDimID
    GROUP BY cp.ClientName, c.CategoryName
) ranked
WHERE rnk = 1;"

dbGetQuery(mart, sql)
```

```
##                     ClientName CategoryName TotalSpentUSD rnk
## 1                    BHP Group       Travel      191274.6   1
## 2                     Glencore       Travel      200071.4   1
## 3                    Rio Tinto       Travel      205385.0   1
## 4 Southern Copper Corporation       Travel      194250.6   1
## 5                      Vale SA       Travel      191956.2   1
```

Currency-Based

Question 8: Which currency has the most transactions?

```r
sql <- "
SELECT
    cu.CurrencyName,
    COUNT(*) AS TransactionCount
FROM ExpenseFacts f
JOIN CurrencyDim cu ON f.CurrencyDimID = cu.CurrencyDimID
GROUP BY cu.CurrencyName
ORDER BY TransactionCount DESC
LIMIT 5"

dbGetQuery(mart, sql)
```

```
##   CurrencyName TransactionCount
## 1          EUR              783
## 2          BRL              772
## 3          AUD              758
## 4          USD              750
## 5          GBP              738
```

Billable Facts Report

GENERAL TABLE

```r
sql <- "
SELECT *
FROM BillableFacts
LIMIT 5"

dbGetQuery(mart, sql)
```

```
##   BillableFactsID DateDimID ClientProjectDimID EmployeeDimID Year Quarter
## 1               2      3002                  1             1 2025       4
## 2               8      3001                  1             1 2025       4
## 3               9      4001                  1             1 2025       4
## 4              12      2001                  1             1 2025       3
## 5              13      2003                  1             1 2025       3
##   BillableAmountUSD NonBillableAmountUSD BillableCount NonBillableCount
## 1           7763.17              1451.13            61               10
## 2           8122.63              1236.40            60               12
## 3           8333.65              2377.99            54               21
## 4            246.05               215.79             2                1
## 5            663.42                38.33             6                1
##   TotalAmountUSD
## 1        9214.29
## 2        9359.03
## 3       10711.64
## 4         461.84
## 5         701.74
```

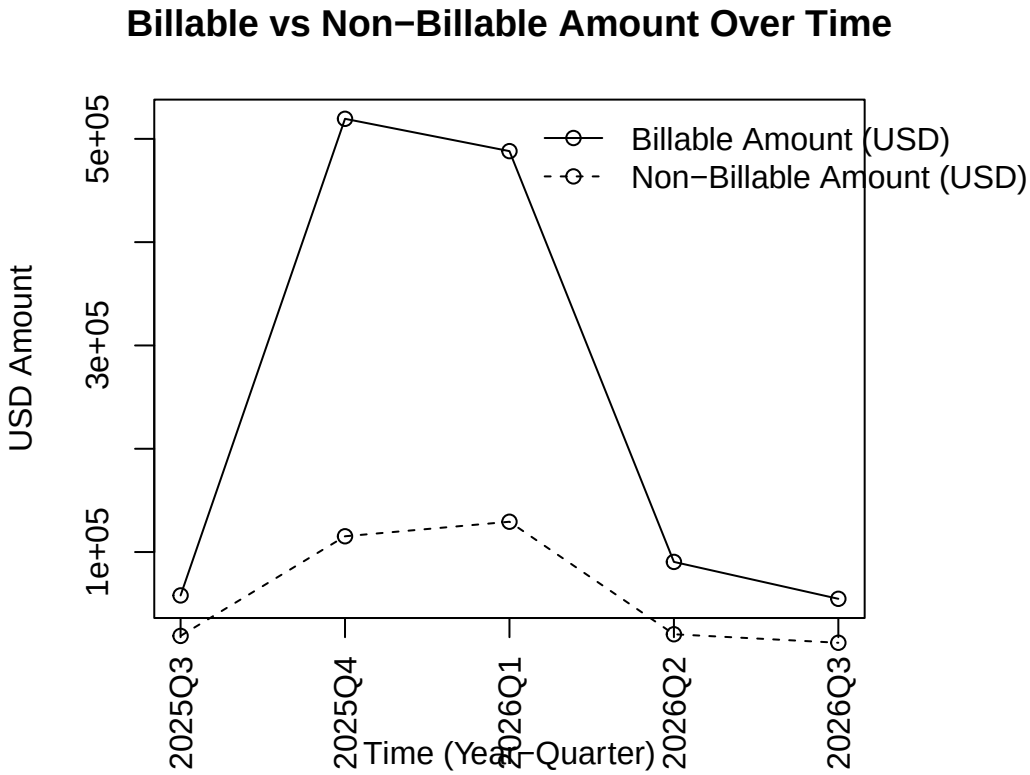
Question 1: What is the total billable vs non-billable amount over time?

```r
sql <- "
SELECT
    f.Year,
    f.Quarter,
    SUM(f.BillableAmountUSD) AS TotalBillableUSD,
    SUM(f.NonBillableAmountUSD) AS TotalNonBillableUSD
FROM BillableFacts f
GROUP BY f.Year, f.Quarter
ORDER BY f.Year, f.Quarter"

df <- dbGetQuery(mart, sql)
df
```

```
##   Year Quarter TotalBillableUSD TotalNonBillableUSD
## 1 2025       3         58004.38            18835.00
## 2 2025       4        519407.28           115265.26
## 3 2026       1        488121.44           129331.26
## 4 2026       2         90460.74            20415.32
## 5 2026       3         54732.48            12184.76
```

```r
# Create time label
df$Time <- paste(df$Year, "Q", df$Quarter, sep="")
df$Time <- factor(df$Time, levels = df$Time)

# Expand right margin for legend
par(mar = c(5, 4, 4, 10), xpd = TRUE)

# Plot Billable
plot(
  x = seq_along(df$Time),
  y = df$TotalBillableUSD,
  type = "o",
  xaxt = "n",
  xlab = "Time (Year-Quarter)",
  ylab = "USD Amount",
  main = "Billable vs Non-Billable Amount Over Time"
)

# X-axis labels
axis(1, at = seq_along(df$Time), labels = df$Time, las = 2)

# Add Non-Billable line
lines(
  x = seq_along(df$Time),
  y = df$TotalNonBillableUSD,
  type = "o",
  lty = 2
)

# Legend on the right
legend(
  "topright",
  inset = c(-0.25, 0),
```

```
  legend = c("Billable Amount (USD)", "Non-Billable Amount (USD)"),
  lty = c(1, 2),
  pch = 1,
  bty = "n"
)
```



Question 2: Which quarter has the highest billable revenue?

```
sql <- "
SELECT
    f.Year,
    f.Quarter,
    SUM(f.BillableAmountUSD) AS TotalBillableUSD
FROM BillableFacts f
GROUP BY f.Year, f.Quarter
ORDER BY TotalBillableUSD DESC
LIMIT 1"
```

Question 3: Which employees generate the highest billable revenues?

```
sql <- "
SELECT
    e.EmployeeName,
    SUM(f.BillableAmountUSD) AS TotalBillableUSD
FROM BillableFacts f
JOIN EmployeeDim e
    ON f.EmployeeDimID = e.EmployeeDimID
GROUP BY e.EmployeeName
ORDER BY TotalBillableUSD DESC;"
```

BudgetVsActual Facts Report

GENERAL TABLE

```r
sql <- "
SELECT *
FROM BudgetVsActualFacts
LIMIT 5"

dbGetQuery(mart, sql)
```

```
##   BudgetFactsID DateDimID ClientProjectDimID Year Quarter ProjectBudget
## 1             5      4001                  1 2025       4         5e+05
## 2             7      2003                  1 2025       3         5e+05
## 3             8      3002                  1 2025       4         5e+05
## 4            11      3001                  1 2025       4         5e+05
## 5            13      2001                  1 2025       3         5e+05
##   ActualSpendUSD Variance
## 1       20662.64 479337.4
## 2        1546.18 498453.8
## 3       19451.85 480548.2
## 4       19951.53 480048.5
## 5        1480.95 498519.0
```

Question 1: Which projects have the worst budget accuracy?

```r
sql <- "
SELECT
    cp.ProjectName,
    SUM(f.ProjectBudget) AS TotalBudget,
    SUM(f.ActualSpendUSD) AS TotalActual,
    SUM(f.Variance) AS TotalVariance,
    (SUM(f.Variance) / NULLIF(SUM(f.ProjectBudget), 0)) * 100 AS VariancePercent
FROM BudgetVsActualFacts f
JOIN ClientProjectDim cp
    ON f.ClientProjectDimID = cp.ClientProjectDimID
GROUP BY cp.ProjectName
ORDER BY ABS(VariancePercent) DESC
LIMIT 5"

dbGetQuery(mart, sql)
```

```
##                       ProjectName TotalBudget TotalActual TotalVariance
## 1           BHP Compliance Review    11250000    149350.4      11100650
## 2      Glencore Compliance Review    10500000    151404.1      10348596
## 3 Southern Copper Site Assessment     9000000    150785.2       8849215
## 4          Vale Compliance Review     8250000    147136.2       8102864
## 5             BHP Site Assessment     7500000    150126.5       7349874
##   VariancePercent
## 1        98.67244
## 2        98.55806
## 3        98.32461
## 4        98.21653
## 5        97.99831
```

Question 2: What is the variance trend year-over-year?

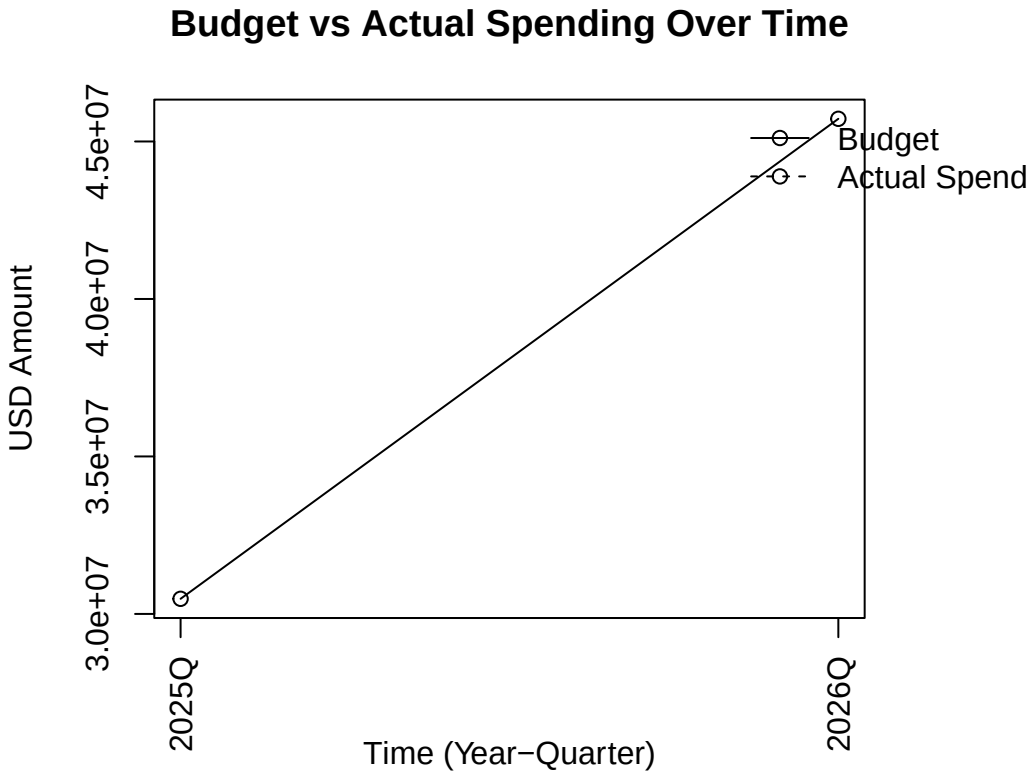
```
sql <- "
SELECT
    f.Year,
    SUM(f.ProjectBudget) AS TotalBudget,
    SUM(f.ActualSpendUSD) AS TotalActual,
    SUM(f.Variance) AS TotalVariance,
    (SUM(f.Variance) / NULLIF(SUM(f.ProjectBudget), 0)) * 100 AS VariancePercent
FROM BudgetVsActualFacts f
GROUP BY f.Year
ORDER BY f.Year
LIMIT 10"

df <-dbGetQuery(mart, sql)
df
```

```
##   Year TotalBudget TotalActual TotalVariance VariancePercent
## 1 2025    30480000      711512      29768488        97.66564
## 2 2026    45720000      795246      44924754        98.26062
```

GRAPH

```
# Time label
df$Time <- paste(df$Year, "Q", df$Quarter, sep="")
df$Time <- factor(df$Time, levels = df$Time)

# Expand margin for legend
par(mar = c(5, 4, 4, 10), xpd = TRUE)

# Plot Budget
plot(
  x = seq_along(df$Time),
  y = df$TotalBudget,
  type = "o",
  xaxt = "n",
  xlab = "Time (Year-Quarter)",
  ylab = "USD Amount",
  main = "Budget vs Actual Spending Over Time"
)

# X-axis labels
axis(1, at = seq_along(df$Time), labels = df$Time, las = 2)

# Add Actual line
lines(
  x = seq_along(df$Time),
  y = df$TotalActual,
  type = "o",
  lty = 2
)

# Legend on the side
legend(
  "topright",
  inset = c(-0.25, 0),
```

```
  legend = c("Budget", "Actual Spend"),
  lty = c(1, 2),
  pch = 1,
  bty = "n"
)
```



Question 3: What is the variance trend year and quarter?

```r
sql <- "
SELECT
    f.Year,
    f.Quarter,
    SUM(f.ProjectBudget) AS TotalBudget,
    SUM(f.ActualSpendUSD) AS TotalActual,
    SUM(f.Variance) AS TotalVariance,
    (SUM(f.Variance) / NULLIF(SUM(f.ProjectBudget), 0)) * 100 AS VariancePercent
FROM BudgetVsActualFacts f
GROUP BY f.Year, f.Quarter
ORDER BY f.Year, f.Quarter
LIMIT 5"

dbGetQuery(mart, sql)
```

```
##   Year Quarter TotalBudget TotalActual TotalVariance VariancePercent
## 1 2025       3    15240000    76839.48      15163161        99.49580
## 2 2025       4    15240000   634672.53      14605327        95.83548
## 3 2026       1    15240000   617452.66      14622547        95.94847
## 4 2026       2    15240000   110876.09      15129124        99.27247
## 5 2026       3    15240000    66917.26      15173083        99.56091
```

```r
dbDisconnect(mart)
```

```
## [1] TRUE
```